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Course: DBMS

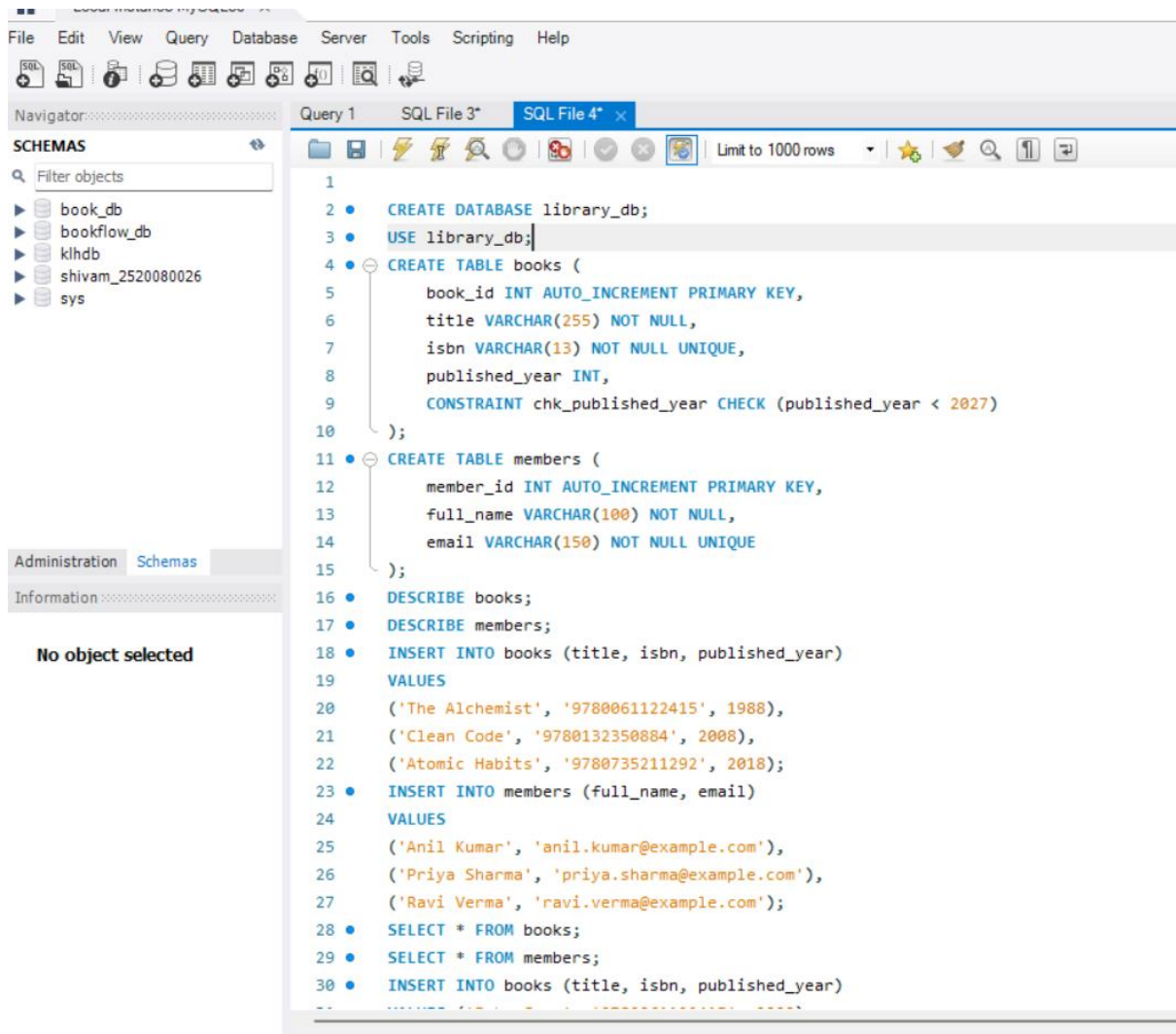
Practical week 1&2

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## **Practical**

### **Week 1:**

A local library is currently managing its books and members using a physical notebook, which has resulted in several operational problems such as misplaced books, difficulty tracking borrowed items, and inaccurate member records. To modernize its operations, the library plans to develop a database system called BookFlow. As a database developer, your task is to create the SQL foundation for this system. First, create a new database named `bookflow_db`. Then, design two tables: a `Books` table containing the columns `book_id`, `title`, `isbn`, and `published_year`, and a `Members` table containing the columns `member_id`, `full_name`, and `email`. Ensure data integrity by setting `book_id` and `member_id` as PRIMARY KEYS, applying a NOT NULL constraint to the `title` column so that every book must have a name, enforcing UNIQUE constraints on both `isbn` and `email` to prevent duplicate entries, and adding a CHECK constraint to ensure that the `published_year` is not in the future (for example, it must be less than 2027). Finally, insert three sample records into both the `Books` and `Members` tables to verify that the database structure and constraints are functioning correctly. This exercise will help you develop skills in schema definition, data integrity enforcement, and the use of SQL Data Definition Language (DDL) commands such as `CREATE DATABASE`, `CREATE TABLE`, and `INSERT` statements.



OUTPUT:

| book_id | title                 | isbn          | published_year |
|---------|-----------------------|---------------|----------------|
| 1       | To Kill a Mockingbird | 9780061120084 | 1960           |
| 2       | 1984                  | 9780451524935 | 1949           |
| 3       | The Hobbit            | 9780547928227 | 1937           |

| #  | Time     | Action                                                                                                                                                                                                       | Message                                                               | Duration / Fetch |
|----|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|------------------|
| 2  | 15:09:00 | CREATE DATABASE library_db                                                                                                                                                                                   | 1 row(s) affected                                                     | 0.031 sec        |
| 3  | 15:09:07 | USE library_db                                                                                                                                                                                               | 0 row(s) affected                                                     | 0.016 sec        |
| 4  | 15:09:07 | CREATE TABLE Books ( book_id INT PRIMARY KEY AUTO_INCREMENT, title VARCHAR(255) NOT NULL, isbn VARCHAR(13) NOT NULL UNIQUE, published_year INT, CONSTRAINT chk_published_year CHECK (published_year < 2027)) | 0 row(s) affected                                                     | 0.031 sec        |
| 5  | 15:09:07 | CREATE TABLE Members ( member_id INT PRIMARY KEY AUTO_INCREMENT, full_name VARCHAR(100) NOT NULL, email VARCHAR(150) NOT NULL UNIQUE)                                                                        | 0 row(s) affected                                                     | 0.015 sec        |
| 6  | 15:09:07 | INSERT INTO Books (title, isbn, published_year) VALUES ('To Kill a Mockingbird', '9780061120084', 1960), ('1984', '9780451524935', 1949), ('The Hobbit', '9780547928227', 1937)                              | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0                | 0.016 sec        |
| 7  | 15:09:07 | INSERT INTO Members (full_name, email) VALUES ('Asha Rao', 'asha.rao@example.com'), ('Vikram Singh', 'vikram.singh@example.com'), ('Anil Kumar', 'anil.kumar@example.com')                                   | 3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0                | 0.000 sec        |
| 8  | 15:11:24 | CREATE TABLE books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, isbn VARCHAR(13) NOT NULL UNIQUE, published_year INT, CONSTRAINT chk_published_year CHECK (published_year < 2027)) | Error Code: 1050. Table 'books' already exists                        | 0.000 sec        |
| 9  | 15:11:39 | CREATE TABLE books ( book_id INT AUTO_INCREMENT PRIMARY KEY, title VARCHAR(255) NOT NULL, isbn VARCHAR(13) NOT NULL UNIQUE, published_year INT, CONSTRAINT chk_published_year CHECK (published_year < 2027)) | Error Code: 1050. Table 'books' already exists                        | 0.000 sec        |
| 10 | 15:12:05 | CREATE DATABASE library_db                                                                                                                                                                                   | Error Code: 1007. Can't create database 'library_db': database exists | 0.000 sec        |

## Practical week 2

The Librarian's Dashboard Scenario The library is now digital! The head librarian needs a report to see which members are active and which books are in the collection. You also need to handle a "Book Donation" process where a new book is added and assigned to a category at the exact same time—if one part fails, the whole

process must stop to keep the catalog accurate. Task to be Done 1. TheCatalog Search (JOINS): Create a third table called Loans that connects member\_idto book\_id. Write a query to show the Member Name and the Book Title they havecurrently borrowed. 2. Collection Stats (Aggregations): Write a query to find thetotal number of books published in each year. o Hint: Use COUNT(book\_id) andGROUP BY published\_year. 3. The Donation Transaction (ACID): Use aTransaction (BEGIN...COMMIT) to add a new book to the Books table andsimultaneously log the entry in a Donation\_History table. 4. Search Speed(Indexing): Create an Index on the isbn column. Explain why searching by ISBN isnow faster for the librarian than searching by Title. Skill Set Gained • RelationalLogic: Connecting independent tables (Books + Members) through a Junction table(Loans). • Analytical SQL: Using GROUP BY to generate business insights. •Atomic Operations: Using Transactions to ensure multiple database changessucceed or fail as a single unit.

The screenshot shows the MySQL Workbench interface for a local instance of MySQL 8.0. The 'SCHEMAS' pane on the left lists databases: book\_db, bookflow\_db, klhdb, shivam\_2520080026, and sys. The 'Administration' tab is selected, and the 'Schemas' sub-tab is active. The 'Information' pane at the bottom shows 'No object selected'. The main editor displays a SQL script with the following queries:

```

1 • CREATE DATABASE skytrack_db;
2 • USE skytrack_db;
3 -- Flights table
4 • CREATE TABLE Flights (
5     flight_id      INT PRIMARY KEY,
6     flight_number  VARCHAR(10) NOT NULL UNIQUE,
7     source         VARCHAR(50) NOT NULL,
8     destination    VARCHAR(50) NOT NULL,
9     departure_date  DATE NOT NULL,
10    ticket_price    DECIMAL(10,2) CHECK (ticket_price > 0)
11 );
12 -- Passengers table
13 • CREATE TABLE Passengers (
14     passenger_id   INT PRIMARY KEY,
15     passenger_name VARCHAR(100) NOT NULL,
16     email          VARCHAR(100) UNIQUE
17 );
18 -- Bookings table (junction table)
19 • CREATE TABLE Bookings (
20     booking_id     INT PRIMARY KEY,
21     passenger_id   INT NOT NULL,
22     flight_id      INT NOT NULL,
23     booking_date   DATE NOT NULL,

```

Local instance MySQL80 x

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- book\_db
- bookflow\_db
- klhdb
- shivam\_2520080026
- sys

Administration Schemas

Information

No object selected

Query 1 SQL File 3\* SQL File 4\* SQL File 5\* x

Limit to 1000 rows

```
25 FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
26 );
27
28 -- Flight_History table (for the transaction task)
29 CREATE TABLE Flight_History (
30     history_id INT PRIMARY KEY,
31     flight_id INT NOT NULL,
32     action_type VARCHAR(50) NOT NULL,
33     action_date DATE NOT NULL,
34     FOREIGN KEY (flight_id) REFERENCES Flights(flight_id)
35 );
36 -- Insert 10 flights
37 INSERT INTO Flights (flight_id, flight_number, source, destination, departure_date, ticket_price) VALUES
38 (1,'AI101','Hyderabad','Delhi','2026-09-01',4500.00),
39 (2,'AI102','Delhi','Mumbai','2026-09-01',5200.00),
40 (3,'AI103','Mumbai','Bengaluru','2026-09-02',3800.00),
41 (4,'AI104','Bengaluru','Chennai','2026-09-02',3200.00),
42 (5,'AI105','Chennai','Hyderabad','2026-09-03',2900.00),
43 (6,'AI106','Hyderabad','Kolkata','2026-09-03',5600.00),
44 (7,'AI107','Kolkata','Delhi','2026-09-04',6100.00),
45 (8,'AI108','Delhi','Bengaluru','2026-09-04',5900.00),
46 (9,'AI109','Mumbai','Hyderabad','2026-09-05',3300.00),
47 (10,'AI110','Chennai','Mumbai','2026-09-05',4100.00);
```

File Edit View Query Database Server Tools Scripting Help

Navigator

Filter objects

- book\_db
- bookflow\_db
- klhdb
- shivam\_2520080026
- sys

Administration Schemas

Information

No object selected

Query 1 SQL File 3\* SQL File 4\* SQL File 5\* x

Limit to 1000 rows

```
46 (9,'AI109','Mumbai','Hyderabad','2026-09-05',3300.00),
47 (10,'AI110','Chennai','Mumbai','2026-09-05',4100.00);
48 -- Insert 10 passengers
49 INSERT INTO Passengers (passenger_id, passenger_name, email) VALUES
50 (201,'Arjun Reddy','arjun.reddy@example.com'),
51 (202,'Sneha Kapoor','sneha.kapoor@example.com'),
52 (203,'Rohit Verma','rohit.verma@example.com'),
53 (204,'Divya Menon','divya.menon@example.com'),
54 (205,'Karan Malhotra','karan.malhotra@example.com'),
55 (206,'Ananya Iyer','ananya.iyer@example.com'),
56 (207,'Farhan Sheikh','farhan.sheikh@example.com'),
57 (208,'Neha Joshi','neha.joshi@example.com'),
58 (209,'Vivek Nair','vivek.nair@example.com'),
59 (210,'Ritika Sharma','ritika.sharma@example.com');
60 -- Insert 12 bookings (10+ records)
61 INSERT INTO Bookings (booking_id, passenger_id, flight_id, booking_date) VALUES
62 (1,201,1,'2026-08-01'),(2,202,2,'2026-08-01'),(3,203,3,'2026-08-02'),
63 (4,204,4,'2026-08-02'),(5,205,5,'2026-08-03'),(6,206,6,'2026-08-03'),
64 (7,207,7,'2026-08-04'),(8,208,8,'2026-08-04'),(9,209,9,'2026-08-05'),
65 (10,210,10,'2026-08-05'),(11,201,6,'2026-08-06'),(12,202,7,'2026-08-06');
66 -- Display all tables
67 SELECT * FROM Flights;
68 SELECT * FROM Passengers;
```

```

67 • SELECT * FROM Flights;
68 • SELECT * FROM Passengers;
69 • SELECT * FROM Bookings;
70 -- INNER JOIN: passenger name, flight number, source, destination
71 • SELECT p.passenger_name AS Passenger_Name, f.flight_number AS Flight_Number,
72         f.source AS Source, f.destination AS Destination
73 FROM Bookings b
74 INNER JOIN Passengers p ON b.passenger_id = p.passenger_id
75 INNER JOIN Flights f ON b.flight_id = f.flight_id;
76 -- GROUP BY: total flights per destination
77 • SELECT destination, COUNT(flight_id) AS Total_Flights
78 FROM Flights
79 GROUP BY destination
80 ORDER BY destination;
81 -- Transaction: add flight + log to history atomically
82 • START TRANSACTION;
83 • INSERT INTO Flights (flight_id, flight_number, source, destination, departure_date, ticket_price)
84 VALUES (11, 'AI111', 'Hyderabad', 'Pune', '2026-09-10', 3600.00);
85 • INSERT INTO Flight_History (history_id, flight_id, action_type, action_date)
86 VALUES (1, 11, 'NEW_FLIGHT_ADDED', CURDATE());
87 • COMMIT;
88 -- Index on flight_number
89 • CREATE INDEX idx_flight_number ON Flights(flight_number);

```

OUTPUT:

| flight_id | flight_number | source    | destination | departure_date | ticket_price |
|-----------|---------------|-----------|-------------|----------------|--------------|
| 1         | AI101         | Hyderabad | Delhi       | 2026-09-01     | 4500.00      |
| 2         | AI102         | Delhi     | Mumbai      | 2026-09-01     | 5200.00      |
| 3         | AI103         | Mumbai    | Bengaluru   | 2026-09-02     | 3800.00      |
| 4         | AI104         | Bengaluru | Chennai     | 2026-09-02     | 3200.00      |
| 5         | AI105         | Chennai   | Hyderabad   | 2026-09-03     | 2900.00      |
| 6         | AI106         | Hyderabad | Kolkata     | 2026-09-03     | 5600.00      |
| 7         | AI107         | Kolkata   | Delhi       | 2026-09-04     | 6100.00      |
| 8         | AI108         | Delhi     | Bengaluru   | 2026-09-04     | 5900.00      |
| 9         | AI109         | Mumbai    | Hyderabad   | 2026-09-05     | 3300.00      |
| 10        | AI110         | Chennai   | Mumbai      | 2026-09-05     | 4100.00      |
| NULL      | NULL          | NULL      | NULL        | NULL           | NULL         |

| Result Grid    |                |                            |  |
|----------------|----------------|----------------------------|--|
| Filter Rows:   |                | Edit:                      |  |
| Export/Import: |                | Wrap Cell Content:         |  |
| passenger_id   | passenger_name | email                      |  |
| 201            | Arjun Reddy    | arjun.reddy@example.com    |  |
| 202            | Sneha Kapoor   | sneha.kapoor@example.com   |  |
| 203            | Rohit Verma    | rohit.verma@example.com    |  |
| 204            | Divya Menon    | divya.menon@example.com    |  |
| 205            | Karan Malhotra | karan.malhotra@example.com |  |
| 206            | Ananya Iyer    | ananya.iyer@example.com    |  |
| 207            | Farhan Sheikh  | farhan.sheikh@example.com  |  |
| 208            | Neha Joshi     | neha.joshi@example.com     |  |
| 209            | Vivek Nair     | vivek.nair@example.com     |  |
| 210            | Ritika Sharma  | ritika.sharma@example.com  |  |
| NULL           | NULL           | NULL                       |  |

| Result Grid    |              |                    |              |  |
|----------------|--------------|--------------------|--------------|--|
| Filter Rows:   |              | Edit:              |              |  |
| Export/Import: |              | Wrap Cell Content: |              |  |
| booking_id     | passenger_id | flight_id          | booking_date |  |
| 1              | 201          | 1                  | 2026-08-01   |  |
| 2              | 202          | 2                  | 2026-08-01   |  |
| 3              | 203          | 3                  | 2026-08-02   |  |
| 4              | 204          | 4                  | 2026-08-02   |  |
| 5              | 205          | 5                  | 2026-08-03   |  |
| 6              | 206          | 6                  | 2026-08-03   |  |
| 7              | 207          | 7                  | 2026-08-04   |  |
| 8              | 208          | 8                  | 2026-08-04   |  |
| 9              | 209          | 9                  | 2026-08-05   |  |
| 10             | 210          | 10                 | 2026-08-05   |  |
| 11             | 201          | 6                  | 2026-08-06   |  |
| 12             | 202          | 7                  | 2026-08-06   |  |
| NULL           | NULL         | NULL               | NULL         |  |

| Result Grid    |               |                    |             |  |
|----------------|---------------|--------------------|-------------|--|
| Filter Rows:   |               | Export:            |             |  |
|                |               | Wrap Cell Content: |             |  |
| Passenger_Name | Flight_Number | Source             | Destination |  |
| Arjun Reddy    | AI101         | Hyderabad          | Delhi       |  |
| Arjun Reddy    | AI106         | Hyderabad          | Kolkata     |  |
| Sneha Kapoor   | AI102         | Delhi              | Mumbai      |  |
| Sneha Kapoor   | AI107         | Kolkata            | Delhi       |  |
| Rohit Verma    | AI103         | Mumbai             | Bengaluru   |  |
| Divya Menon    | AI104         | Bengaluru          | Chennai     |  |
| Karan Malhotra | AI105         | Chennai            | Hyderabad   |  |
| Ananya Iyer    | AI106         | Hyderabad          | Kolkata     |  |
| Farhan Sheikh  | AI107         | Kolkata            | Delhi       |  |
| Neha Joshi     | AI108         | Delhi              | Bengaluru   |  |
| Vivek Nair     | AI109         | Mumbai             | Hyderabad   |  |
| Ritika Sharma  | AI110         | Chennai            | Mumbai      |  |

| Result Grid |             |               | Filter Rows: | Export: | Wrap Cell Content: |
|-------------|-------------|---------------|--------------|---------|--------------------|
|             | destination | Total_Flights |              |         |                    |
| ▶           | Bengaluru   | 2             |              |         |                    |
|             | Chennai     | 1             |              |         |                    |
|             | Delhi       | 2             |              |         |                    |
|             | Hyderabad   | 2             |              |         |                    |
|             | Kolkata     | 1             |              |         |                    |
|             | Mumbai      | 2             |              |         |                    |

| Result Grid |           |               |         |             |                |              | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |
|-------------|-----------|---------------|---------|-------------|----------------|--------------|--------------|-------|----------------|--------------------|
|             | flight_id | flight_number | source  | destination | departure_date | ticket_price |              |       |                |                    |
| ▶           | 5         | AI105         | Chennai | Hyderabad   | 2026-09-03     | 2900.00      |              |       |                |                    |
| *           | NULL      | NULL          | NULL    | NULL        | NULL           | NULL         |              |       |                |                    |